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Question 11

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Final Exam due Jan 21, 2025 04:09 IST Completed

Question 11

1.0/1.0 point (graded)

Consider Dijkstra’s algorithm for finding the shortest path between two nodes in a graph with negative costs. Which of the following statements regarding the convergence of the algorithm is true?

- ☐ Solution 1. The algorithm will perform at most as many iterations as their are nodes in the graph.
- ☒ Solution 2. If no cycle with negative costs exists, the algorithm will terminate in a finite number of steps.
- ☐ Solution 3. The algorithm will always terminate in a finite number of steps.
- ☐ Solution 4. The algorithm can stop once the destination node has been visited.



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